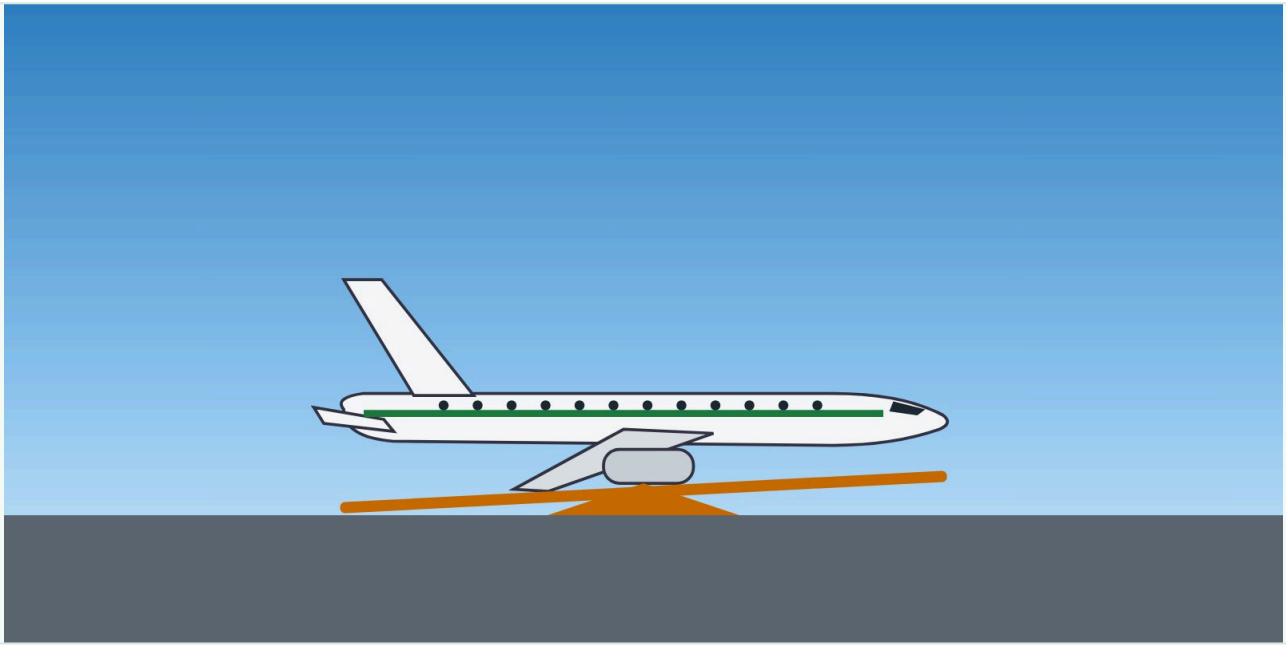
MODULE 3 · CERTIFICATION, PERFORMANCE AND LOADING

Lesson 11: Weight and Balance Effects

Lesson 11 of 24 · about 7 min

LEARNING OBJECTIVES

- Explain how weight affects stall speed, performance and buffet margin.
- Describe the effects of forward and aft centre of gravity on stability and control.
- Recognise how loading errors can contribute to an upset.

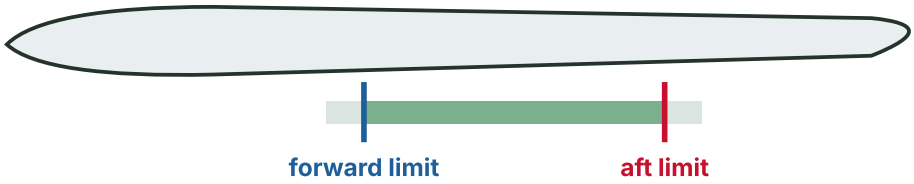


Weight and centre of gravity shape stability, stall speed and control authority.

Weight

- Stall speed increases with the square root of weight. A heavier aircraft must fly faster for the same margin.
- At high altitude, higher weight reduces the buffet margin and the maximum altitude (Lesson 7).
- Climb performance and acceleration are lower at high weight — less energy is available in a recovery.
- An incorrect weight entered in the FMS produces incorrect speeds, which can erode the stall margin from take-off onwards.

Centre of gravity



Forward CG
more stable, heavier pitch forces,
higher stall speed, more elevator to rotate and flare

Aft CG
less stable, lighter pitch forces,
easy to over-control, less nose-down authority

The CG must stay within its certified limits. Loading errors outside the envelope can leave too little control authority to prevent or recover from an upset.

	Forward CG	Aft CG
Longitudinal stability	Greater	Reduced
Pitch control forces	Heavier	Lighter — easy to over-control
Stall speed	Slightly higher	Slightly lower
Elevator authority	More needed to rotate and flare	Less nose-down authority for stall recovery
Fuel consumption	Slightly higher (more tail down-force)	Slightly lower

Loading errors

- Cargo or passengers loaded differently from the load sheet can move the CG outside the envelope.
- Unsecured cargo that shifts during rotation or in turbulence can cause a sudden, large pitch change.
- Incorrect trim settings for take-off — from a wrong CG — can cause an early or late rotation and a nose-high or tail-strike situation.

PREVENTION
Check the load sheet and the FMS entries carefully, cross-check weight and CG with the fuel and passenger figures, and be alert to unusual control forces at rotation. If the aircraft does not feel right, say so.

KEY TAKEAWAYS

- Higher weight: higher stall speed, lower buffet margin, less excess energy.
- Aft CG: less stable, lighter forces, less nose-down authority. Forward CG: heavier forces, higher stall speed.
- Loading and FMS weight errors erode margins before the flight begins.

KNOWLEDGE CHECK

What is a typical effect of an aft centre of gravity?

A. Heavier pitch control forces

B. Reduced longitudinal stability and lighter pitch forces

C. Higher stall speed

D. More nose-down elevator authority

Answer: B. An aft CG reduces stability and makes pitch forces lighter, so the aircraft is easier to over-control.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.